Bruce Cheng

AI Application Engineer / AI Solution Builder
bruce.zheng0814@gmail.com | github.com/bruce-lcz | linkedin.com/in/bruce-lcz

Summary

AI Application Engineer focused on turning ambiguous enterprise needs into usable AI workflows and internal products. Experienced across on-premise LLM serving, RAG/agent workflows, OCR, ASR, document intelligence, manufacturing decision support, and workflow automation. I work between requirement framing, system composition, backend implementation, and user-facing delivery, with a strong emphasis on security-constrained environments and practical team adoption.

Experience

PixArt Imaging Inc. Aug 2025 – Present
Information Engineer (Generative AI / LLM Projects) *Hsinchu, Taiwan*

- Build on-premise Generative AI workflows for Legal, R&D, engineering, and internal knowledge scenarios in a semiconductor / IC design environment with strict data-security constraints.
- **Internal GenAI Workspace:** Designed and integrated a shared internal workspace around vLLM model serving, document/image utilities, and task-oriented interaction flows, helping engineering teams adopt AI for coding support, document review, and technical analysis.
- **ASC Merge & C Code Generator:** Productionized a manual ASC merge and C initialization-code conversion process into an Open WebUI tool using Python native tool calling, raw-byte decoding fallback, Docker Compose, and Nginx-hosted downloadable artifacts.
- **Legal Contract Comparison System:** Built an offline document comparison pipeline combining OCR, LLM-based paragraph restructuring, multiset validation, and diff review to reduce slow, error-prone manual contract checks.
- **Patent Translation Workflow:** Designed a local LLM workflow for first-draft patent translation with staged prompting, proofreading support, terminology consistency handling, and DOCX export for sensitive IP documents.
- **Auto-Minutes Meeting Intelligence:** Engineered an offline meeting workflow combining ASR, speaker-aware segmentation, text cleanup, and LLM summarization to turn internal audio into structured minutes and summaries.

AUO Corporation - Advanced Manufacturing Center Aug 2021 – Aug 2025
AI Engineer / Project Leader *Taichung, Taiwan*

- Led manufacturing-facing AI projects that connected models, data pipelines, explainability, and internal tools to real engineering and operational decision workflows.
- **Manufacturing Assistant Platform (RAG → Agent):** Evolved internal documentation retrieval into a domain-aware assistant capable of retrieval, SQL querying, API/tool use, structured diagnostics, and process-knowledge reasoning.
- **Cross-Factory Yield Optimization:** Developed a Golden Path optimization workflow and LIME-based explainable analysis to translate cross-factory variability into actionable engineering parameters, improving yield by 1.43%, reducing defects by 12.62%, and saving approx. 300 man-hours monthly.
- **Preventive Maintenance Workflow:** Designed an LSTM-based time-series prediction workflow with a custom trend-consistency objective, reducing defect prediction RMSE by 20% and saving approx. 90 man-hours monthly.
- Recognized as Smart Manufacturing Level 3 Elite Talent; preventive maintenance outcomes were approved for an internal defensive patent.

Academia Sinica - GIS Center Feb 2019 – Jul 2019
Research Intern *Taipei, Taiwan*

- Built a data quality workflow for parsing and normalizing ambiguous Taiwan address strings across heterogeneous research datasets.
- Developed rule-based normalization logic, structured address attributes, and a web-based sampling/review interface; won the Best Student Paper Award at TGIS 2019.

Selected Projects

ASC Merge & C Code Generator | *Open WebUI, Python, Native Tool Calling, Docker, Nginx* 2026

- Converted a repetitive SD configuration-file workflow into a guided internal tool supporting single-file conversion, ordered multi-file merge, legacy encoding fallback, comment preservation, and downloadable C/ASC outputs.

Internal GenAI Workspace | *Python, vLLM, Model Serving, Document Utilities, Docker* 2025 – Present

- Combined internal model access, reusable interaction patterns, and document/image utilities into a shared workspace for engineering productivity and recurring AI-assisted workflows.

Legal Contract Comparison System | *Python, FastAPI, vLLM, OCR, Human Review Workflow* 2025 – Present

- Implemented an offline document-intelligence pipeline for contract comparison, using OCR, LLM restructuring, omission checks, and diff visualization to support safer legal review.

Manufacturing Assistant Platform | *RAG, LangGraph, SQL, API Tools, Knowledge Graph* 2024 – 2025

- Expanded an internal manufacturing assistant from document retrieval into a tool-using decision-support workflow for diagnostics, structured queries, and process reasoning.

Private AI Gym Coach | *React, TypeScript, Cloudflare Workers, D1, LLM API* 2026 – Present

- Built a personal fitness product around repeated workout-tracking behavior, using AI selectively for exercise mapping and coaching prompts while keeping the core UI fast, private, and workflow-focused.

Skills

AI Workflow & LLM Systems: On-Premise LLM Serving, vLLM, RAG, Agent Workflows, Tool Calling, LangGraph, Knowledge Graph, Prompt Engineering, Validation/Fallback Design

Document, Speech & Multimodal AI: OCR, Document Parsing, Patent/Legal Workflows, ASR, Speaker Segmentation, Vision-LLM, Meeting Intelligence

Backend & Product Engineering: Python, FastAPI, Django, REST APIs, Microservices, Docker, Cloudflare Workers, Cloudflare D1, AWS Bedrock, Workflow Orchestration

Data Science & Manufacturing AI: Time-Series Forecasting, LSTM, Custom Loss Functions, Explainable AI (LIME), Genetic Algorithms, SQL, Databricks, ETL

Frontend & Internal Tools: React, TypeScript, Streamlit, Tailwind CSS, Internal Tool UX, Git, CI/CD, Linux

Education

National Yunlin University of Science and Technology 2019 – 2021
Master's Degree, Industrial Engineering and Management *Yunlin, Taiwan*

- **Thesis:** Improve the Predicted Efficiency of Crowd Flow in Large Areas Based on the R-tree and the SPACE-MDL-LSTM

National Yunlin University of Science and Technology 2015 – 2019
Bachelor's Degree, Industrial Engineering and Management *Yunlin, Taiwan*

- **Thesis:** Optimization of Machine Dispatching in Flexible Production Processes

Awards

Smart Manufacturing Level 3 Elite Talent | *AUO Corporation* 2025

Best Student Paper Award | *Taiwan Geographic Information Society Conference (TGIS)* 2020

Best Student Paper Award | *Taiwan Geographic Information Society Conference (TGIS)* 2019